

BLDAP 2025 Coding Survey

The purpose of this coding survey is for the instructors to understand everyone's coding knowledge before starting our introduction to Python/data science/machine learning course. This is not a test and responses are collected anonymously. We (the instructors) want to make sure our intro to Python/data science/machine learning course is the most beneficial for everyone!

We encourage everyone to try their best in answering these questions. **For questions you do not know the answer to, please do not randomly guess and instead choose the option "I am not familiar with this concept."** It is completely fine to choose this option for any or even all questions.

* Indicates required question

1. 4.35 is an example of what data type? *

Mark only one oval.

- ☐ Integer
- ☐ Float
- ☐ Boolean
- ☐ String
- ☐ I am not familiar with this concept.

2. How would you assign the integer value 23 to a variable named student_numbers? *

Mark only one oval.

- ☐ "student_numbers" = 23
- ☐ student_numbers = 23
- ☐ student_numbers = "23"
- ☐ student_numbers == 23
- ☐ I am not familiar with this concept.

3. What would the following code output? *

```
name = "BLDAP"  
len(name)
```

Mark only one oval.

- ☐ "BLDAP"
- ☐ 4
- ☐ 5
- ☐ True
- ☐ I am not familiar with this concept.

4. What is the difference between lists and dictionaries in Python? *

Mark only one oval.

- ☐ Lists are mutable, dictionaries are immutable.
- ☐ Lists use indices to access values, dictionaries use keys to access values.
- ☐ Lists are created with { } ("curly brackets"), dictionaries are created using [] ("square brackets").
- ☐ There are no differences.
- ☐ I am not familiar with this concept.

5. What would the following code output? *

```
temperature_data = [65, 68, 73, 78, 88]  
temperature_data[0:2]
```

Mark only one oval.

- ☐ [65, 68, 73]
- ☐ [65, 68]
- ☐ [73, 78, 88]
- ☐ [78, 88]
- ☐ I am not familiar with this concept.

6. How would you retrieve the value 'Seoul' from the following? *

```
capitals = {  
    'Colombia' : 'Bogota',  
    'Germany' : 'Berlin',  
    'South Korea' : 'Seoul',  
}
```

Mark only one oval.

- ☐ capitals[2]
- ☐ capitals[3]
- ☐ capitals['South Korea']
- ☐ capitals['Seoul']
- ☐ I am not familiar with this concept.

7. What would the following code output? *

```
light_value = 400

if light_value < 200:
    print('It is dim.')
elif light_value < 500:
    print('It is light.')
elif light_value < 800:
    print('It is bright.')
else:
    print('It is VERY bright.')
```

Mark only one oval.

- ☐ It is dim.
- ☐ It is light.
- ☐ It is bright.
- ☐ It is VERY bright.
- ☐ I am not familiar with this concept.

8. What would the following code output? *

```
for i in range(4):  
    print(i)
```

Mark only one oval.

- ☐ 0, 1, 2, 3
- ☐ 0, 1, 2, 3, 4
- ☐ 1, 2, 3, 4
- ☐ i
- ☐ I am not familiar with this concept.

9. What would the following code output? *

```
x = 0  
while x < 5:  
    print(x)  
    x = x / 2
```

Mark only one oval.

- ☐ 0, 0, 0, 0
- ☐ 0, 0, 0, 0, 0
- ☐ x, x, x, x, x
- ☐ A lot of 0... it is an infinite loop!
- ☐ I am not familiar with this concept.

10. What kind of numPy array does the following code create? *

```
test_array = np.zeros((2,5))
```

Mark only one oval.

- ☐ A 2D array with 2 rows, 5 columns and filled with zeros.
- ☐ A 2D array with 5 rows, 2 columns and filled with zeros.
- ☐ A 1D array with 2 rows, filled with fives.
- ☐ A 1D array with 5 rows, filled with twos.
- ☐ I am not familiar with this concept.

11. Which function from the pandas library imports data and stores it into a dataframe? *

Mark only one oval.

- ☐ import()
- ☐ read_csv()
- ☐ to_csv()
- ☐ import_csv()
- ☐ I am not familiar with this concept.

12. What is the difference between loc and iloc in selecting data in pandas dataframes?

Mark only one oval.

- ☐ loc is used for rows only, iloc is used for columns only
- ☐ loc can select multiple rows and columns, iloc can select only one row or column
- ☐ The starting index for loc is 1, while the starting index for iloc is 0
- ☐ loc selects based on row and column labels, iloc selects based on row and column indices
- ☐ I am not familiar with this concept.

13. What information will the following code tell you about the rna_data dataframe? *

```
rna_data.isnull().sum()
```

Mark only one oval.

- ☐ The sum of all values in the dataframe.
- ☐ The code will return 'True' if there are missing values in the dataframe.
- ☐ The number of missing values that exist in the dataframe.
- ☐ The row labels that contain missing values in the dataframe.
- ☐ I am not familiar with this concept.

14. The image shows the dataframe named `star_data`, which contains data on each star's class type and mass relative to the sun. Below the dataframe is a line of code. What will it return? *

	Name	Class	Mass Relative to Sun
0	Sun	G	1.00
1	Vega	A	2.10
2	Sirius	A	2.00
3	TRAPPIST-1	M	0.09
4	Wolf 359	M	0.11
5	Altair	A	1.79
6	Polaris Aa	F	5.40

```
star_data[star_data['Mass Relative to Sun'] > 1.0]
```

Mark only one oval.

- ☐ It will return 'True' as the dataframe contains stars that have a relative mass greater than 1.
- ☐ It will return a list of boolean values with all stars that have a relative mass greater than 1.
- ☐ It will return a list of indices with all stars that have a relative mass greater than 1.
- ☐ It will return a dataframe with all stars that have a relative mass greater than 1.
- ☐ I am not familiar with this concept.

15. On the same dataframe `star_data`, how can you group stars by their class type and find the average mass of stars in each class type? *

Mark only one oval.

- ☐ Select just the column 'Class' with `loc`, then apply the `mean()` function on the column
- ☐ Use the function `groupby()` on the column 'Class', then use the `mean()` function
- ☐ Use the `mean()` function on the whole dataframe, as it will automatically find the average mass by class type
- ☐ Use the `describe()` function on the whole dataframe, as one of the information it returns is the mean
- ☐ I am not familiar with this concept.

16. Which type of plot would be the best choice to show the average mass of stars by class type? *

Mark only one oval.

- ☐ A line plot with the class type as the x-axis and the average masses as the y-axis.
- ☐ A histogram of the stars' masses.
- ☐ A bar plot with the class type as the x-axis and the height of the bars corresponding to the average masses.
- ☐ A pie chart of the stars' masses with each slice representing a class type.
- ☐ I am not familiar with this concept.

17. Which Python library is best suited to visualize data? *

Mark only one oval.

- ☐ matplotlib
- ☐ numPy
- ☐ pandas
- ☐ tensorflow
- ☐ I am not familiar with this concept.

18. What is important to keep in mind when training machine learning algorithms? *

Mark only one oval.

- ☐ You should train the algorithm on as much data as possible, regardless of quality.
- ☐ Machine learning algorithms are not susceptible to biases that might be present in the data.
- ☐ The more complex the algorithm, the less data you need for training.
- ☐ The training data should be like real-world examples the algorithm will face later, not just perfect examples.
- ☐ I am not familiar with this concept.

19. A medical team would like to predict the risk of patients developing a certain disease based on factors such as age, weight, and family health history. What type of machine learning algorithm should the medical team use? *

Mark only one oval.

- ☐ Regression
- ☐ Classification
- ☐ Clustering
- ☐ Dimensionality Reduction
- ☐ I am not familiar with this concept.

20. When building a machine learning model, we split the data into training and testing sets (train test split). Why do we do this? *

Mark only one oval.

- ☐ To organize the data and make it easier to analyze.
- ☐ To reduce the amount of data the model needs to process, making it run faster.
- ☐ To ensure the model can make accurate predictions on completely new, unseen data.
- ☐ Splitting the training data prevents the machine learning model from feeling overwhelmed and lets it learn at its own pace.
- ☐ I am not familiar with this concept.

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